16 The straight line **L** has the equation $3y = 4x + 7$
The point A has coordinates $(3, -5)$

Find an equation of the straight line that is perpendicular to **L** and passes through A.

$3y = 4x + 7 \rightarrow y = (4/3)x + 7/3$

Gradient of L = 4/3.

1 mark
Gradient of L = 4/3.

Perpendicular gradient = −3/4.

1 mark
Perpendicular gradient −3/4.

Through A(3, −5): $y + 5 = -3/4(x - 3)$

$4y + 20 = -3x + 9$

$3x + 4y = -11$

1 mark
Line through (3, −5).

$3x + 4y = -11$

(Total for Question 16 is 3 marks)